| Meghana Rayudu – PES2UG24CS398| Section – G |Ronak Thamaran – PES2UG24CS410 |

Generating Music with Machine Learning

Problem and objective

Melodia ML generates a symbolic piano-MIDI continuation from a short seed phrase. It compares four approaches: a count-based conditional Naive Bayes baseline, a feed-forward neural network, an LSTM, and a GRU encoder-decoder. The project uses a reproducible preprocessing and generation pipeline.

Dataset and implementation

The corpus contains 24 classical-piano MIDI pieces selected from the ClassicalPianoMIDI-dataset, which sources files from Piano-midi.de. Exact source URLs are recorded in data/DATASET_SOURCES.md. MIDI files are parsed into simultaneous pitch groups and rests; durations are quantized to 0.25 quarter-length steps, long pieces are capped at 2,500 events, and each piece is transposed toward C using a duration-weighted pitch-class estimate. A seeded, whole-piece split assigns 20 pieces to training and 4 to the held-out set; the vocabulary is built from training pieces. Original MIDI files are not modified.

Models

Model	Prediction setup	Trainable parameters
Naive Bayes	Smoothed next-event distribution from the previous 3 events	Count-based
Feed-forward NN	40-event context predicts the next event	468,290
LSTM	Recurrent 40-event context predicts the next event	281,890
GRU encoder-decoder	Encodes 40 events; trained to predict an 8-event continuation	297,602

Training and results

The three neural models used the production preset: 500 epochs, context length 40, batch size 64, AdamW with learning rate 0.0005, and gradient clipping. Naive Bayes fits conditional counts once. The recorded metrics in artifacts/metrics.json are:

Model	Recorded accuracy	Recorded cross-entropy
Naive Bayes	7.45%	Not reported
Feed-forward NN	5.62%	2.4141
LSTM	7.54%	2.2687
GRU encoder-decoder	2.40%	2.2971

Evaluation limitation: The current training script evaluates the same four-piece held-out split after every epoch and uses its loss to select neural checkpoints. Accuracy is recorded from the final epoch, while cross-entropy is the minimum observed across epochs; these values may describe different checkpoints. Treat them as exploratory diagnostics, not an independent final test result. Encoder-decoder accuracy uses an 8-event prediction horizon and is not directly comparable to single-next-event accuracy. A stronger evaluation should select checkpoints on a validation split and evaluate once on an untouched test split.

Output and conclusion

Each model produces a standard MIDI file in artifacts/generated/. The renderer preserves model-sampled pitches and durations, checks MIDI pitch range, and adds velocity variation; it does not add a hand-authored melody or suppress repeated notes. In the recorded run, the LSTM has the highest reported next-event accuracy and lowest reported cross-entropy among the neural models, but the evaluation limitation above means this does not establish a definitive winner or prove musical quality. Listening tests should assess harmony, rhythm, phrasing, and repetition.

Reproducibility

```
python -m pip install -r requirements.txt
python -m src.preprocess
python -m src.train_all --preset production --device auto
python -m src.generate --model lstm --events 320 --temperature 0.78 --top-k 12
```

Source: ClassicalPianoMIDI-dataset, https://github.com/cheriell/ClassicalPianoMIDI-dataset. Confirm source terms before redistributing MIDI files. See data/DATASET_SOURCES.md for file-level provenance and PROJECT_GUIDE.md for full setup and methodology.